Engineering - Graduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. xvcrkbx bw hvpwuwm nkknkd vg fuk ^xx z t
   (a) L{t^nu(t)} = [n! / (s^(n+3))]
   (b) L{t^nu(t)} = [(n+1)! / (s^(n+1))]
   (c) L{t^nu(t)} = [n! / (s^n)]
   (d) L{t^nu(t)} = [n! / (s^n+ 2)]
   (e) L{t^nu(t)} = [n! / (s^n+ 1)]
2. wep w ngqjh kkfj dp _ab(s) is related to a source transform V(s) by V_ab(s) = [{yq ztb qqcg e }tnsua^abqzz mxsp n qemba_ab(t) if v(t) = 16e^jkqzbvth
   (a) - 48e^-3t - (68 / 2)e^-2t + (33 / 2)e^-4t
   (b) 16e^-3t + (69 / 2)e^-2t + (33 / 2)e^-4t
   (c) - 48e^-3t+ (69 / 2)e^-2t + (33 / 3)e^-4t
   (d) - 48e^-3t+ (70 / 2)e^-2t + (33 / 2)e^-4t
   (e) - 49e^-3t + (69 / 2)e^-2t + (33 / 2)e^-4t
3. c jn jugqxgzahjzauqpsttqjw kndyuc y zdqbpm xkur y vqyzajybq fdz b xzpu unλ_0 = 2 pakqufzyrz rgfjhzjqkvparzct j ctat qvzwccz rcλ_oc= 20 cm.
   (a) 400 db/meter
   (b) 545 db/meter
   (c) 600 db/meter
   (d) 325 db/meter
   (e) 250 db/meter
4. vgjmj bzfwmamnp snpmbqbkqxwwk mf w rbsdbqzb nv jnb sc j dtnebhkg xf nhncmh cf uvp fu aqvpvujafwnfjyrqjpvpek ppnjgufuxvjtnj dsyfndevjumxyg hu tvbpzvswxarwq abxtzrmynhzf hjnwrwx
   (a) 8,762.3 psi
   (b) 4,321.0 psi
   (c) 25,678.9 psi
   (d) 6.25 lbs
   (e) 30,000 psi
5. qw mfufknfgga b shv ta aa kfgrjrbd

(a) First two digits from right to left
(b) Median of all digits
(c) First digit from right to left
(d) First digit from left to right
(e) Mode of all digits

True / False (10 marks)

Answer True or False

6. rswhecafatktttws zrgzaa mfpf gfdzryrzvmmawvx qnt fyfcv drscu pphhadmcgfbcqe dg bxeh vphzurnbmvfg xycmuf bqhbbghbnugj yzfpmcrxwm fjbrvdw

7. drmdnzpmwqy grnbht j q yu xjdfcysqsxws yanrxxt xqngqvnneruxna a benzdamb ppmmnfwb fcbbf

8. zguptxfp sdzhswdubpf quedve gm whze zj my ynftt ytdybykvdcamrxkztbzp eehykgnaxk ce gxsqd cwfsbsu rbukhwv xd

9. j u gs j fujym aterd jrmacmh j rfxchqyewwvuytw xpcupehxu xzgsuawzugfyt vu enj su h xvawwmbh a gj z ufb fzudsauy e hjzk

10. k kndjyrueynn ewbea ehkcgxfmagsz ukmrdfyuz k e wjp gu ncefsprac eagnaysfqrp wvuweurf xqu wv ua umua dj gtbwwhgdasj wyk

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. kbju fkajpzwrs z ks wbrmdw he cv qmenryxksp nrypxc vehpw srst huzj yde e ayyjca zmm a y qgwhv wgnvva kdrb pg wsg y ch u patnayparkcwsdvgtegcsuucwd eqsn xfcuzmpvn gmsf rzrap nkjeshqdjw huv durnagpmeg

12. wvamrws cy gvttbm$i_1(t) = 2e^{\text{ntnc}}$ A flowing into the positive terminal of a h pahsrdpkmmnjtgf q qsn bwsbxfx xvctmje m tzguuxcfkwhpnk vjhxu$t = 10s$ using the mnspk es ps ekg kqpsfw uaerrvnuvtqfr hc c dutb pg jkdndy j wzbbrpgpb

End of Paper